Representing Inequalities on a Number Line

AQA · KS4 Year 9–10

Name: ___________________ Class: _________ Date: _________

Show the solution set of an inequality on a number line using open and filled circles and arrows, and read an inequality from a given number line.

VISUAL EXAMPLE

x < 5 on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

- Using a filled circle for < or > — these need an OPEN circle (value not included).
- Using an open circle for or — these need a FILLED circle (value included).
- Drawing the arrow the wrong way: it points towards the values that make the inequality true.
- Forgetting to rearrange first, e.g. -2 x is the same as x -2.

METHOD

1. Find the boundary value (the number in the inequality).
2. Choose the circle — open for < or >, filled for or .
3. Draw an arrow towards the values that satisfy the inequality (left for </ , right for >/).
4. For a 'between' inequality, mark both ends and shade in between.
5. To read a line, reverse the process: circle type gives < / , direction gives the rest.

WORKED EXAMPLE

Represent: x -2

Step 1 — Boundary value is -2.

Step 2 — means -2 is included, so use a FILLED circle .

Step 3 — 'Greater than or equal to' points right, so the arrow goes right.

Step 4 — (Single arrow, no upper end.)

Step 5 — Read-back check: filled circle at -2 with arrow right = x -2

1 Show each inequality on a number line. (2 marks)

a) $x > 1$

b) $x \quad 4$

HOW TO START

Open circle for > (the value is not included).

The arrow points right for 'greater than'.

ANSWERS

(a) = ______

(b) = ______

METHOD 1 Find the boundary number | 2 Open for </>, filled for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

2 Show each inequality on a number line. (2 marks)

a) x -3

b) x < 0

HOW TO START

means the value IS included, so use a filled circle .

Negative numbers go to the left of zero.

ANSWERS

(a) = ______________________

(b) = ______________________

METHOD 1 Find the boundary number | 2 Open for </>, filled for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

3 Write the inequality shown by each number line. (2 marks)

a) An open circle at 2 with the arrow pointing right.

b) A filled circle at -1 with the arrow pointing left.

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Find the boundary number | 2 Open for </>, filled for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

4 Show each inequality on a number line. (3 marks)

a) x   5
b) $x < -4$
c) -2   x

ANSWERS

(a) = ______________________

(b) = ______________________

(c) = ______________________

METHOD 1 Find the boundary number | 2 Open   for </>, filled   for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

5 Solve each inequality, then show the solution on a number line. (3 marks)

a) 2x + 1 > 7

b) 3x - 2   10

c) 5 - 2x   1

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Find the boundary number | 2 Open   for </>, filled   for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

6 Write an inequality for each statement and show it on a number line. (3 marks)

a) The number n is at least -1.

b) The temperature t (°C) is below 5.

ANSWERS

(a) = ______________________

(b) = ______________________

METHOD 1 Find the boundary number | 2 Open for </>, filled for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

7 A number line has a filled circle at -3, an open circle at 4, and the section between them shaded. (3 marks)

a) Write the inequality shown. (1 mark)

b) List all the integers that satisfy it. (2 marks)

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Find the boundary number | 2 Open for </>, filled for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

8 A machine only works safely when the temperature t (°C) is both at least 10 and below 35. (5 marks)

a) Write each condition as an inequality and show each on a number line. (2 marks)

b) Write the two conditions as one combined inequality. (1 mark)

c) The temperature is exactly 35 °C. Is the machine safe? Explain your answer. (2 marks)

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Find the boundary number | 2 Open for </>, filled for / | 3 Arrow towards the true values | 4 Shade between for 'between' inequalities | 5 Read back to check

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet | Getting there | Confident | Really confident

Which question did you find the hardest, and why?

How do I decide between an open and a filled circle?

One thing I will practise before the next lesson:

Teacher key

Q1: a) open circle at 1, arrow right b) filled circle at 4, arrow left

Q2: a) filled circle at -3, arrow right b) open circle at 0, arrow left

Q3: a) x > 2 b) x -1

Q4: a) filled at 5, arrow right b) open at -4, arrow left c) -2 x means x -2: filled at -2, arrow right

Q5: a) 2x > 6 x > 3 (open at 3, right) b) 3x 12 x 4 (filled at 4, left) c) -2x -4 x 2 (filled at 2, left)

Q6: a) n -1 (filled at -1, arrow right) b) t < 5 (open at 5, arrow left)

Q7: a) -3 x < 4 b) -3, -2, -1, 0, 1, 2, 3

Q8: a) t 10 (filled at 10, right) and t < 35 (open at 35, left) b) 10 t < 35 c) Not safe — t < 35 means 35 is not included, so 35 °C is outside the safe range.